

PROJECT CONCEPT NOTE

CyberGuide AI

Intelligent Security Awareness Platform

AI • SECURITY AWARENESS • CLOUD

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Course: Gen AI & Cloud Computing

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LIVE AWS APPLICATION URL: [ADD AFTER DEPLOYMENT]

1. Project Overview

Concept: A responsive AI-powered web application that helps learners understand digital threats, information protection, cloud security, and practical defensive measures through explanations, quizzes, and guided learning.

Item	Details
Application name	CyberGuide AI
Application type	Intelligent Security Awareness Platform
Primary audience	Students, beginners, general internet users, and early-career professionals
Backend	Python FastAPI with streamed HTTP responses
LLM API	Groq API using the Llama 3.3 70B Versatile model
Cloud deployment	Docker container deployed through AWS App Runner

2. Problem Statement

Digital users regularly encounter phishing, scams, weak authentication, insecure cloud practices, malware, and social-engineering attempts. Security guidance is often fragmented or too technical for beginners. This can prevent learners from understanding how threats work and which defensive actions are appropriate.

CyberGuide AI addresses this gap by providing a single, approachable platform where users can explore defined security domains and receive AI-generated explanations adapted to their level.

3. Objective

- Provide simple, accurate explanations of security concepts and internet threats.
- Connect each threat with practical prevention, detection, response, and recovery measures.
- Support active learning through AI-generated quizzes and examples.
- Deliver responses progressively through a streaming interface.
- Demonstrate secure API integration, containerization, and public AWS deployment.

4. Target Users and Use Cases

Item	Details
Students	Revise core concepts, request simple explanations, and generate

Item	Details
	quick quizzes.
General users	Learn how to recognize scams, phishing, malware indicators, and risky online behavior.
Beginners	Select a learning level and receive accessible guidance without excessive jargon.
Early-career users	Explore information, network, and cloud-security fundamentals using practical examples.

5. Key Application Features

- Seven learning domains: Internet Threats, Information Security, Cloud Security, Network Security, Social Engineering, Malware, and Defensive Measures.
- Three learning levels: beginner, intermediate, and advanced.
- Three output modes: concept explanation, defensive measures, and quiz generation.
- Suggested questions and domain cards that prepare focused prompts.
- Real-time streamed AI responses with copy functionality.
- Responsive lilac-themed interface optimized for desktop and mobile.
- Defensive-use guardrails that redirect unsafe requests toward safe learning.

6. Expected User Experience and Outcomes

A user opens the HTTPS application, selects a security domain and learning preference, enters a question, and receives an answer progressively. The interface clearly communicates that the application is intended for defensive education. By the end of a session, the learner should understand the selected threat or concept, recognize common warning signs, and identify realistic protective measures.

7. Success Criteria

- The application loads successfully on desktop and mobile browsers.
- The server keeps the LLM API key outside frontend code and version control.
- AI responses appear progressively and remain focused on safe defensive learning.
- The Docker image starts reliably and the health endpoint reports a healthy status.
- The final AWS URL is publicly accessible over HTTPS at submission time.